



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LAB 14 INSTRUCTIONS

For

Investigate retransmission and zero window

Course Code: C1123

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5.1

Lab 14 — Investigate retransmission and zero window

Course: Wireshark Network Analysis Masterclass (C1123)

Topic: 14 — TCP Protocol Analysis

Learning outcome: LO4 — Diagnose TCP and application behaviour, including HTTP and authorised TLS inspection.

Time: about 60 minutes

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Scenario

A file download stalls part-way through. You must show whether the sender, the receiver or the path held things up.

Goal

Sequence numbers count bytes; ACK numbers indicate the next expected byte.

What you will produce

A TCP evidence sheet for retransmission, zero window and reset.

Before you start

- Wireshark 4.6 or later, with the TShark command-line tools on PATH.
- Python 3 for the verification and export scripts (Windows: use `py -3` in place of `python3`).
- Open this lab folder in a terminal; every file the lab needs is inside it.
- Use only the supplied synthetic captures — do not capture on a network you are not authorised to monitor.

Files for this lab

File	What it is for
data/branch-office.pcap	Synthetic branch-office capture used by the lab
assets/scenario.md	The help-desk ticket that sets the scenario
assets/checks.json	Filters and frame counts the fixture must satisfy
assets/expected-evidence.png	The packet list your filter should produce
scripts/verify.py	Checks the capture facts with TShark
scripts/export_evidence.py	Exports this lab's evidence rows to outputs/evidence.csv
outputs/findings.md	Findings sheet you complete
assets/tcp-evidence.csv	TCP evidence sheet you complete

Lab at a glance

1. Read Expert Information for stream 3
2. Compare the retransmitted segment
3. Confirm the zero-window advertisement
4. Identify the port 81 reset

Step-by-step

1. Prepare your lab workspace. Open this lab folder in a terminal. Windows uses `py -3` in place of `python3`. Wireshark GUI alone is sufficient for the investigation; CLI scripts require TShark on PATH. Run the fixture verification first.

```
python3 scripts/verify.py
```

2. Open `data/branch-office.pcap`. Apply `tcp.stream == 3`. Open Analyze > Expert Information and note the repeated server segment.

3. Apply `tcp.analysis.retransmission` and expand TCP. Compare the same sequence range and payload in the earlier segment; the repeat occurs 1.000 second later.

4. Apply `tcp.window_size_value == 0`. Confirm the client ACK advertises zero receive window in stream 3. Examine the SYN options: window scaling is negotiated, but zero remains zero.

5. Apply `tcp.flags.reset == 1`. Identify the port 81 refusal. Complete `assets/tcp-evidence.csv` with stream, frame, flag/analysis, measured interval and a limitation. Discuss SACK, fast recovery and out-of-order evidence without claiming those events occur in this fixture.

6. Export a reproducible packet table. Record the filter, frame numbers, measurement, explanation and limitation in `outputs/findings.md`.

```
python3 scripts/export_evidence.py
```

Expected evidence

With the lab filter applied, your packet list should match the frames below (produced by TShark from this lab's own capture).

tcp.analysis.retransmission					
No.	Time	Source	Destination	Protocol	Info
51	3.270000	192.0.2.20	192.0.2.10	TCP	[TCP Retransmission] 80 → 51004 [PSH, ACK] Seq=1 ...
branch-office.pcap Displayed: 1					
tcp.analysis.zero_window					
No.	Time	Source	Destination	Protocol	Info
52	3.280000	192.0.2.10	192.0.2.20	TCP	[TCP ZeroWindow] 51004 → 80 [ACK] Seq=70 Ack=93 W...
branch-office.pcap Displayed: 1					
tcp.flags.reset == 1					
No.	Time	Source	Destination	Protocol	Info
56	3.320000	192.0.2.20	192.0.2.10	TCP	81 → 52000 [RST, ACK] Seq=1 Ack=1 Win=8192 Len=0
branch-office.pcap Displayed: 1					

Lab 14 expected evidence

Test it

The supplied tcp.analysis.retransmission expression matches 1 frame(s) in the specified capture. Run scripts/verify.py and compare the listed frame numbers. Keep your findings and exported table in outputs/. Explain the observed result rather than only copying a count.

Troubleshooting

- TShark not found: install Wireshark CLI tools and add the installation folder to PATH; on Windows use the Wireshark install directory.
- A filter returns zero: clear other filters, use the specified capture, and check the expression is in the display toolbar.
- TLS remains opaque: select the matching lab-tls.keys file by absolute path, reload, and remove a key-log preference from another lab.

Try it with TShark

The same evidence from the command line — run it from this lab folder:

```
tshark -n -r data/branch-office.pcap -q -z expert
```

Challenge

Develop and explain an alternative filter.

Reflection

Which second observation point would strengthen your conclusion?

Extension (optional)

TCP lab — find retransmissions and window behaviour in a larger file transfer trace. Source: J.F. Kurose and K.W. Ross, Wireshark Labs (gaia.cs.umass.edu/kurose_ross/wireshark.php).

Reset

Clear all display filters and return to the C1123-Analyst or Default profile. If you loaded a TLS key log, remove it from Preferences > Protocols > TLS. The supplied captures never need regenerating for the core lab.

Note: The same steps appear in the Learner Guide. The slides show only the scenario and a summary.